

Website Development named as 'Cryptness' (Security And Cryptography)

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Abstract — In Today's era, it has become a basic necessity to store our personal details, bank details stored at specific digital location. But, have we ever pondered over the safety and security of our data, which is most valuable – as it can cause various crimes as well as can do deeds of a great importance and use. Here, we introduce the term named as '**Cyber And Data Security**'. This term can be effectively implemented by encrypting the user's data, so that no one can handle or access the data except the verified user. This paper discusses about a website named as – '**Cryptness**' which implements the above process and securely stores the data and fetches it back to user whenever needed by using '**Encryption And Cryptography**'

Keywords — Cryptography, Cyber Security, Data Security, Encryption, SHA-256, Website Development

I. INTRODUCTION

The Greek meaning of Cryptography is Secret Writing. Data Security is one of the major concerns in today's world. Cryptography is a method to fulfill the confidentiality of data. Cryptography provides the privacy of data for an individual or organization by converting the original data to a readable understandable format for an unauthorized person and converting back to original data for the authorized individual[2]. Cryptography Technique translates the plain text to another format, which is called cipher Text with an Encryption Key and can also convert cipher text back to plain text with the same or different encryption key based on the technique used to encrypt the data.

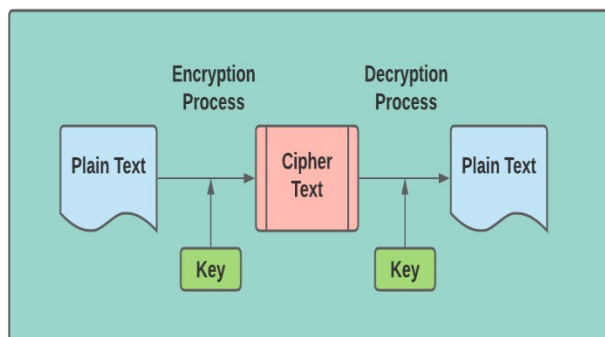


Fig.1. Process Of Cryptography

A. Characteristics Of Cryptography

- Data Security – Secures the plain text
- Data Breaches – Protects from the release of confidential information to the environment
- Authentication And Authorization – Allow only the authenticate individual to read/write the data.

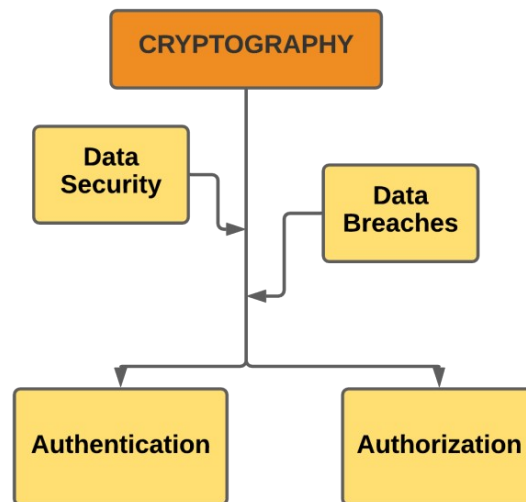


Fig.2. Characteristics of Cryptography

B. Types Of Cryptography

Cryptography is a method that is linked to convert plain text into cipher text i.e called encryption and convert back to plain text i.e known as decryption. Various type of encryption and decryption are available but the most commonly used algorithms for cryptography are :

1) Symmetric Key Cryptography

2) Asymmetric Key Cryptography

In this website we are using, Symmetric Cryptography method.

SYMMETRIC KEY CRYPTOGRAPHY

In this Cryptographic approach, a single key is used to encrypt and decrypt the data. The key must be known to

both the clients to read/write the data. There are different methods of Symmetric key Algorithm like DES, 3DES, AES, and more.[1]

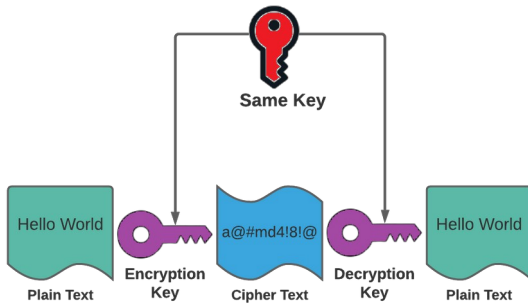


Fig.3. Symmetric Cryptography

The thing we have to note here it that, this website particularly only encrypts the data and do not further decrypts it. This encryption the performed by SHA-256 Hash Function, explained further in Methodology section. The Decryption will act as a disadvantage in our project, as its sole motive is to secure the data and the decryption act as contradiction.

Data security has become most important aspect while transmission of data and storage. The transmission and exchange of image also needs a high security. Cryptography is the art of secret writing. Cryptography is used to maintain security. The basic service provided by cryptography is the ability to send information between participants in a way that prevents others from reading it. Figure 4 shows the information transmitting between sender and receiver. Figure 5 shows the creation of interrupts between sender and receiver. Figure 6 shows the changes, theft or delete information between sender and receiver.

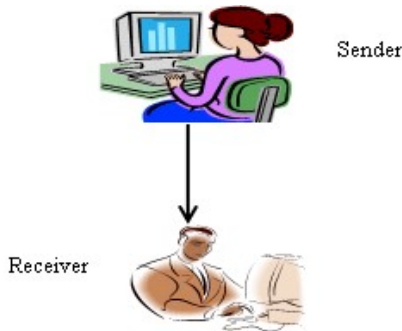


Fig.4. Sender And Receiver Of Information

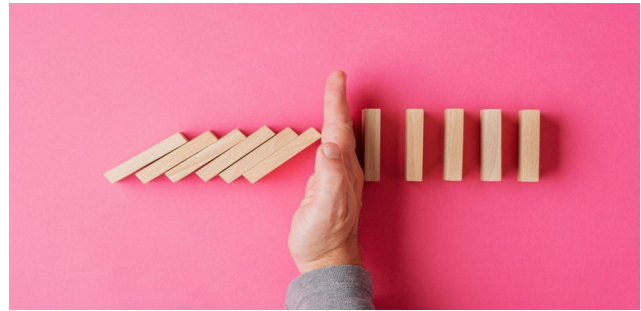


Fig.5. Interruption In Information Exchange



Fig.6. Changes, Theft or Delete Information

Security Standards

1. Architecture
2. Performance
3. Security
4. Flexibility
5. Scalability
6. Limitations

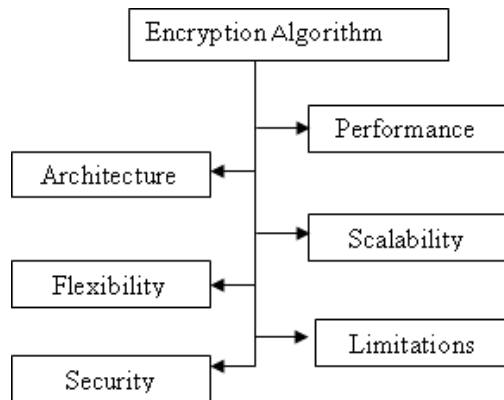


Fig.7. Security Parameters

II. LITERATURE REVIEW

1. Reema Gupta proposed "EFFICIENT ENCRYPTION TECHNIQUES IN CRYPTOGRAPHY BETTER SECURITY ENHANCEMENT." The paper promotes the encryption technique and reviewed their restrictions and approach. Also, talked about the trans-positional approach like Simple columnar, simple row, Route Cipher transposition.[1]
2. Tushar¹, Aniket Sharma², Ankit Mishra³ proposed "CRYPTOGRAPHIC ALGORITHM FOR ENHANCING DATA SECURITY: A THEORETICAL APPROACH". The Greek meaning of Cryptography is Secret Writing. Data Security is one of the major concerns in today's world. Cryptography is a method to fulfill the confidentiality of data. Cryptography provides the privacy of data for an individual or organization by converting the original data to a readable ununderstandable format for an unauthorized person and converting back to original data for the Authorized individual[1]. Cryptography Technique translates the plain text to another format, which is called cipher Text with an Encryption Key and can also convert cipher text back to plain text with the same or different encryption key based on the technique used to encrypt the data.[2]
3. V. Esther Jyothi et al 2020 IOP Conf. Ser.: Mater. Sci. Eng. 981 022028 proposed "ANALYSIS OF CRYPTOGRAPHY ENCRYPTION FOR NETWORK SECURITY". Symmetric Encryption : The same cryptographic keys for plaintext crypting and deterioration of the figure materials are used when Symmetrical Encryption happens. Their only downside is that both clients have to transfer their key's security more quickly. Symmetrical key encryption is less complex. For encryption and decryption of both data, one key is used. Symmetric key types Symmetric-key encryption may use either stream cyphers or block cyphers. Symmetric-key types [4]. Stream cyphers encrypt a message's digits one at a time (typically bytes). Square figures use separate sections and encrypt them with the plaintext as a lone component unit in order to change the measurement of the component. 64-bit squares have been used routinely. The estimation of NIST's Advanced Encryption Standard (AES) the GCM part figure operating method is 128-piece in December 2001.[3]
4. Kristin Bower (A Senior Thesis submitted in partial fulfillment of the requirements for graduation in the Honors Program Liberty University Spring 2019) proposed "CRYPTOGRAPHY: MATHEMATICAL ADVANCEMENTS ON CYBER SECURITY". How does the government keep its secrets from prying eyes? How do companies who promise to keep their customers' personal information safe actually protect that information? In most cases, the answer is cryptography, the study of encoding and decoding information. Starting with rudimentary beginnings, cryptography evolved to include complex mathematical techniques. As mathematicians discovered new aspects of prime numbers and functions with unique properties, the government updated the standard practice for encoding and decoding information in order to keep up with the growing field of mathematics. Still today, researchers seek for new methods to add to cryptography, and as mathematics continues to advance so will the area of cyber security.[4]
5. Jad El-Abed¹, Patryk Szewczyk^{1,2}, 1Edith Cowan University, Perth, Australia 2Security Research Institute, Perth, Australia, p.szewczyk@ecu.edu.au proposed "PERSISTENT ISSUES IN ENCRYPTION SOFTWARE: A HEURISTIC AND COGNITIVE WALKTHROUGH". The support information accompanying security software can be difficult to understand by end-users, who have little knowledge in cyber security. One mechanism for ensuring the integrity and confidentiality of information is encryption software. Unfortunately, software usability issues can hinder an end-user's capability to properly utilize the security features effectively. To date there has been little research in investigating the usability of encryption software and proposing solutions for improving them. This research paper analyzed the usability of encryption software targeting end-users. The research identified several issues that could impede the ability of a novice end-user to adequately utilize the encryption software. A set of proposed recommendations are suggested to improve encryption software which could be empirically verified through further research.[5]

III. METHODOLOGY/EXPERIMENTAL

A. Materials / Components

1. Coding Tools : Python, Python FLASK Framework, YAML
2. Designing Tools : HTML, CSS, JavaScript

B. Objectives

1. A Website Development to store user's data safely using Encryption term from Cryptography.
2. Giving user An Encrypted Private Key as password, using it for further purposes such as - Login, Sign In, Data Storing & Fetching, etc.
3. Encryption protects your privacy.
4. Encryption prevents Identity Theft and Ransomware Blackmail.
5. You Can Use It Across a Variety of Devices
6. It Supports Data Integrity
7. Using Encryption Technology for Data Protection Could Increase Trust

8. Using Encryption Technology for Data Protection Could Increase Trust
9. Avoid coding errors
10. Encryption key services prevent service providers from accessing your data.

C. Encryption

Encryption is the process through which cryptographers transform secure data into code. Early cryptographers used simple methods like substitution (letter A is read as B, B is read as C, and so forth). Today's encryption techniques are much more complicated, requiring the aid of complex computer technology to create and decipher. One way data is encrypted is to convert it into a scrambled code of letters and numbers. Only those who possess the key can decipher the encrypted message.

Data encryption is a way of translating data from plaintext (unencrypted) to cipher text (encrypted). Users can access encrypted data with an encryption key.

There are massive amounts of sensitive information managed and stored online in the cloud or on connected servers. Encryption uses cyber security to defend against brute-force and cyber-attacks, including malware and ransomware. Data encryption works by securing transmitted digital data on the cloud and computer systems. There are two kinds of digital data, transmitted data or in-flight data and stored digital data or data at rest.

Modern encryption algorithms have replaced the outdated Data Encryption Standard to protect data. These algorithms guard information and fuel security initiatives including integrity, authentication, and non-repudiation. The algorithms first authenticate a message to verify the origin. Next, they check the integrity to verify that contents have remained unchanged. Finally, the non-repudiation initiative stops sends from denying legitimate activity.

Symmetric Encryption Methods

Symmetric encryption is a type of encryption where only one secret symmetric key is used to encrypt the plaintext and decrypt the cipher text.

Common symmetric encryption methods :

- Data Encryption Standards (DES): DES is a low-level encryption block cipher algorithm that converts plain text in blocks of 64 bits and converts them to cipher text using keys of 48 bits.
- Triple DES: Triple DES runs DES encryption three different times by encrypting, decrypting, and then encrypting data again.

- Advanced Encryption Standard (AES): AES is often referred to as the gold standard for data encryption and is used worldwide as the U.S. government standard.
- Twofish: Twofish is considered one of the fastest encryption algorithms and is free to use.



Fig.8. Encryption Process

Hashing

Hashing is the process of scrambling raw information to the extent that it cannot reproduce it back to its original form. It takes a piece of information and passes it through a function that performs mathematical operations on the plaintext. This function is called the hash function, and the output is called the hash value/digest.

Hashing has several key uses in computer science. One that perhaps receives the most attention today in a world where cyber security is key is the use of hashing in encryption.

Because hashed strings and inputs are not in their original form, they can't be stolen the way they could be if they are not hashed. If a hacker reaches into a database and finds an original string like "Ajay's wallet ID 19450," they can simply glean, nab or pilfer this information and use it to their advantage, but if they instead find a hash value like "a67b2," that information is completely useless to them, unless they have a key to decipher it. SHA-1, SHA-2, and MD5, are popular cryptographic hashes.

A good hash function for security purposes must be a unidirectional process that uses a one-way hashing algorithm. Otherwise, hackers could easily reverse engineer the hash to convert it back to the original data, defeating the purpose of the encryption in the first place.

To further increase the uniqueness of encrypted outputs, random data could be added to the input of a hash function. This technique is known as "salting" and guarantees unique output even in the case of identical inputs. For example, hackers can guess users' passwords in a database using a rainbow table or access them using a dictionary attack. Some users may share the same password that, if guessed by the hacker, is stolen for all of them. Adding the salt prevents the hacker from accessing these non-unique passwords as each hash value will now be unique, and will stop any rainbow table attacks.

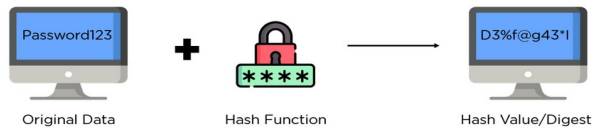


Fig.9. Hashing

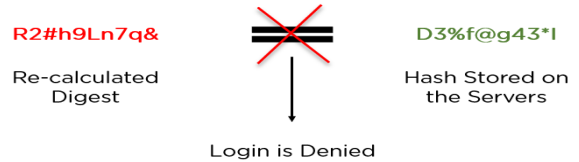


Fig.10. Password Hashes

SHA-256 Cryptographic Hash Algorithm

A cryptographic hash (sometimes called 'digest') is a kind of 'signature' for a text or a data file. SHA-256 generates an almost-unique 256-bit (32-byte) signature for a text. A hash is not 'encryption' – it cannot be decrypted back to the original text (it is a 'one-way' cryptographic function, and is a fixed size for any size of source text). This makes it suitable when it is appropriate to compare 'hashed' versions of texts, as opposed to decrypting the text to obtain the original version.

SHA-256 is one of the successor hash functions to SHA-1 (collectively referred to as SHA-2), and is one of the strongest hash functions available. SHA-256 is one of the most secure and secret hashing functions on the market. The US government agencies use and require this encryption hash function for storing legal documents securely. SHA-256 is built with a Merkle-Damgård structure derived from a one-way compression function itself created with the Davies-Meyer structure from a specialised block cipher.

Working Of SHA-256

The SHA-256 algorithm, like other hash functions, takes any input and produces an output (often called a hash) of fixed length. It doesn't matter if the input is a single word, a full sentence, a page from a book, or an entire book, the output of a hashing algorithm like SHA256 will always be the same length. Specifically, it will be 256 bits, which is 32 bytes, which is displayed as 64 alphanumeric characters. All outputs appear completely random and offer no information about the input that created it.

Other important characteristics of SHA-256 include the fact that it is deterministic (it will always produce the same output when given the same input) and the fact that it is a one-way function. There is no way to reverse engineer an input from knowledge of the output. Lastly, SHA-256 is computationally efficient and an ordinary computer can perform the operation dozens or even hundreds of times per second.

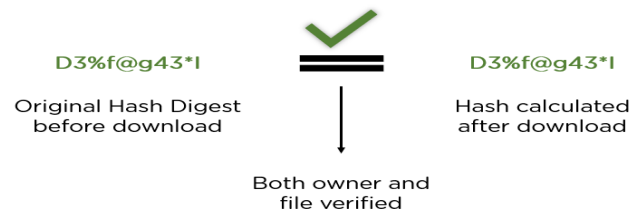


Fig.11. Integrity Verification

Flowchart Of Website (At last of research paper Figure 21)

IV. RESULTS AND DISCUSSIONS

Steps In Algorithm Of SHA-256

1. Padding bits
2. Padding length
3. Initialising buffers
4. Compression functions
5. Output

Step By Step Website Execution Analysis

1. Producing Private Encrypted Key using SHA-256

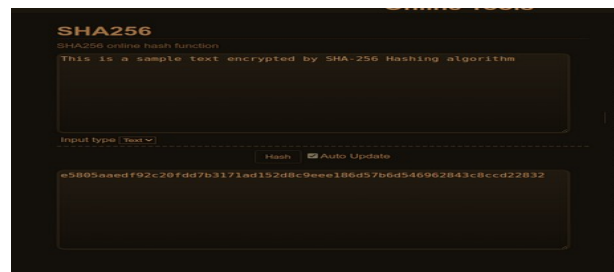


Fig.12. Example Of Encryption By SHA-256 Hash Algorithm

2. Sign In Page

In this webpage, user will create his/her account by filling the details such as : Username, Password and Email ID.

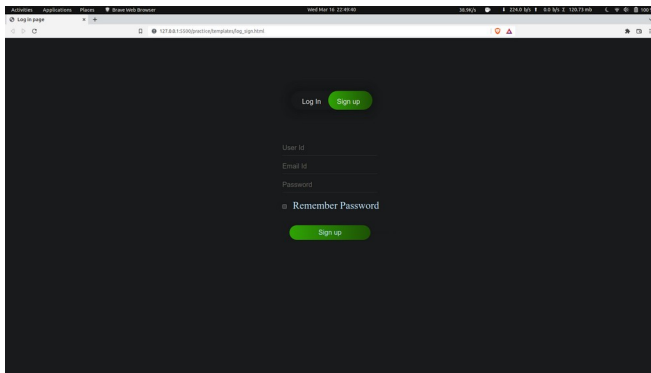


Fig.13. Sign In / Create Account Page

3. Creating Encrypted Private Key And Providing It To User

In this webpage, we will provide the private key to user for 'Security Purposes' and 'Authorization And Authentication Of User'. The key will be provided in the form of 'QR Code' to scan and store purposes. Also a Text of key will be given which can be copied by user.

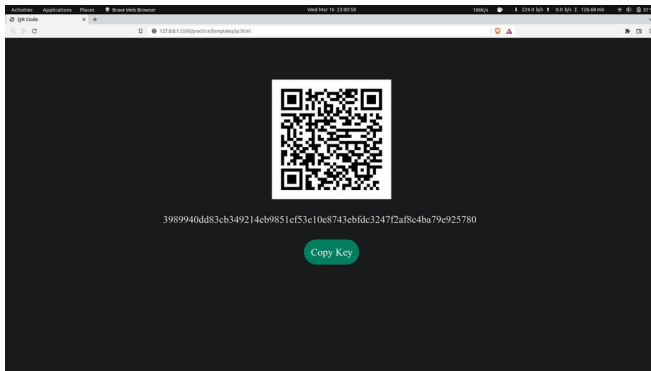


Fig.14. Encrypted Private Key Generation

4. Login Page

Once created account by user, he/she will be directed to Home page where they will have to Login now for further access. The 'Session Concept In Python FLASK' is implemented here to create a session or cookie once the particular user logs in.

This login takes **Password** and given **Encrypted Private Key** to user.

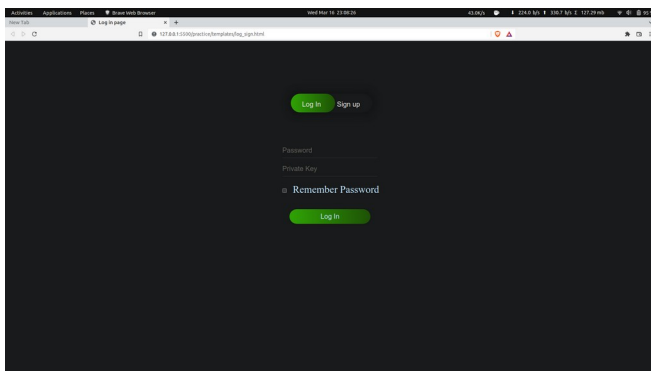


Fig.15. Login Page

After the Login, user will be directed to Home Page.

5. Data Storage Process

Once on the Homepage, user can select to upload the data such as Personal and Bank Details from 'Services' tab.

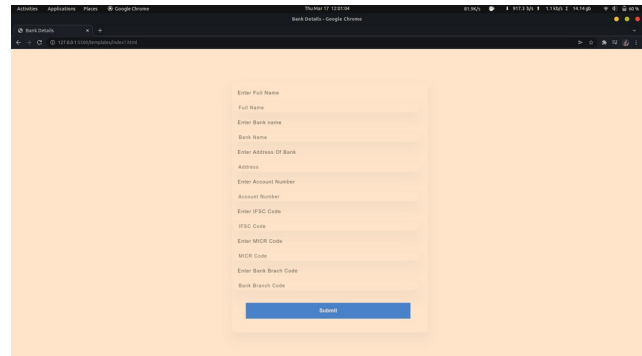


Fig.16. Bank Details Webpage

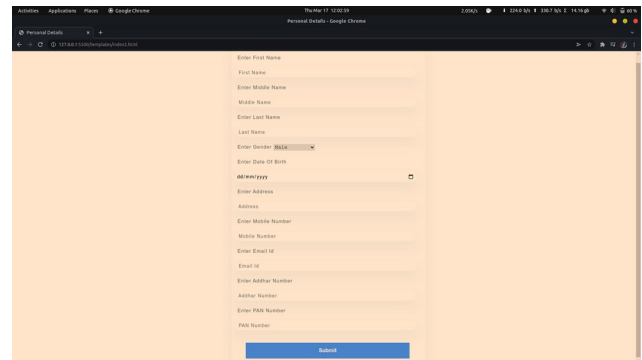


Fig.17. Personal Details Webpage

After filling the form of the desired data type user have to store, the information is stored in a file – each data type (Personal or Bank) has a separate file storing the data of particular user.

6. Data Storage And Encryption

Now, the files that store the data are YAML extensions file. We have used this programming, since YAML is a human-readable data-serialization language. It is commonly used for configuration files and in applications where data is being stored or transmitted. YAML targets many of the same communications applications as Extensible Markup Language but has a minimal syntax and hence easy to handle.

Next, the files are secured as '.zip' extensions files, which furthermore have 'The Encrypted Private Key' as **Password** - given to user while signing in. So, the information cannot be handled or operated by developers as well. The Data is most secure in this case.

7. Fetching The Data Stored

If the user want to fetch the data stored, they will have to input the 'The Encrypted Private Key'. If it matches perfectly with the key stored at Developer end, only and only then the user can access the data. User will be given an option to download the '.zip' extension file.

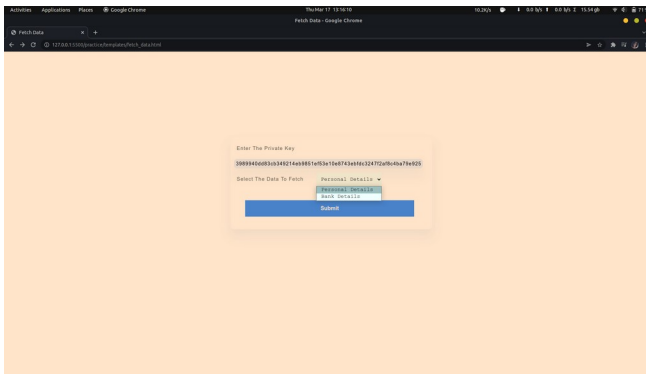


Fig.18. Private Key Verification

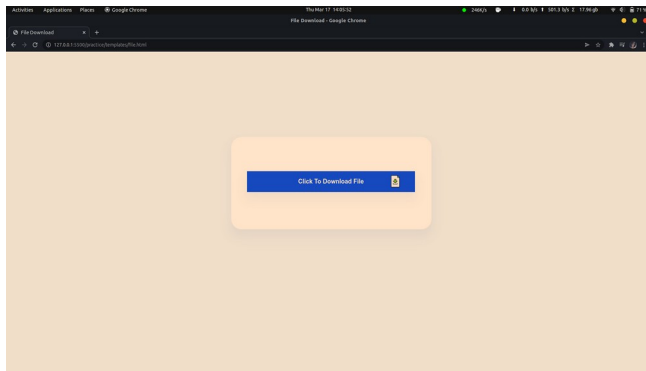


Fig.19. File Download Option Button

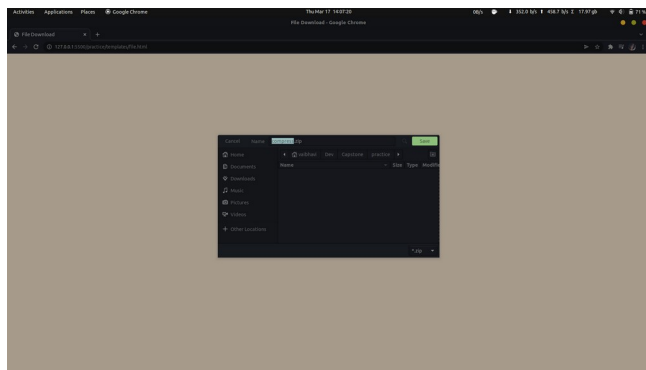


Fig.20. File Saving in User's System

V. CONCLUSION

'Data Encryption Is Privacy Safeguard' can be the conclusion of this project. Data Security and Cyber Security is the need of modern era, which can be skillfully fulfilled by Encryption (SHA-256 Hash Function). It is of great use for securing data from, data hacks, data breaches, trafficking and many more.

Benefits/ Strengths Using Encryption Technology For Data Security

1. Encryption is cheap to implement.
2. Encryption can save you from regulatory fines.
3. Encryption can help to protect remote workers.
4. Encryption increases the integrity of our data.
5. Encryption can increase consumer trust.[7]

Advantages of using SHA-256 encryption for data protection:

1. It is impossible to reconstruct the initial data from hash value.
2. A brute-force attack would need to make 2256 attempts to generate the initial data.
3. Having two or more hash values that are the same is extremely unlikely to happen, hence we can get and generate unique private keys from each user.
4. A minor change to the original data alters the hash value so much that it's not apparent the new hash value is derived from similar data; this is known as the avalanche effect.

So, the minor change in private key and we can get to know that the wrong user is meddling with the data stored and trying to breach it.

ACKNOWLEDGMENT

I am highly indebted to Mr. Amol Ashok Bhosle for their guidance and constant supervision as well as for providing necessary information regarding the project & also for their support in completing the project.

I would like to express my special gratitude and thanks to **Department of Engineering, Sciences and Humanities (DESH)** for implementing the Capstone Project Semester 1 (2021-22 F.Y. B.Tech) in our syllabus. Thanks to DESH Capstone Project Sem-I Coordinator for proper and detailed guidance in project.

My thanks and appreciations also go to my group colleagues in developing the project and people who have willingly helped me out with their abilities.

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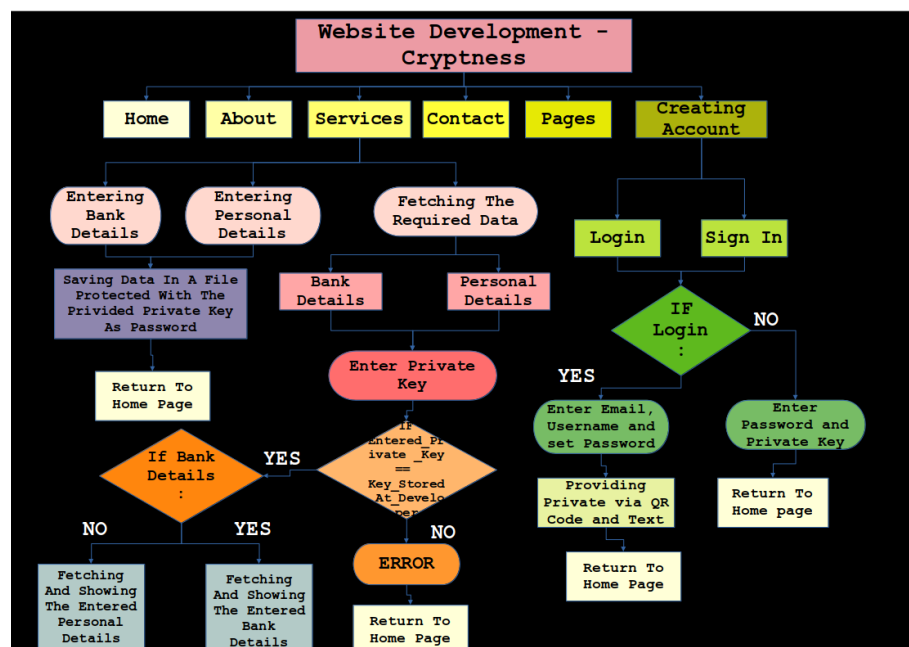


Fig.21. Flowchart of Website